School of Computing and Mathematical Sciences

COMP 1589

Lab week 8

Conditions (If's) in JavaScript

Introduction

In this week's lecture you were told about conditions in JavaScript. In this tutorial you will explore these concepts by creating some simple programs.

Task 1: A simple IF statement

Create a new HTML file called **"week8_task1.html"** and between two `<script> … </script>` tags, type the following code:

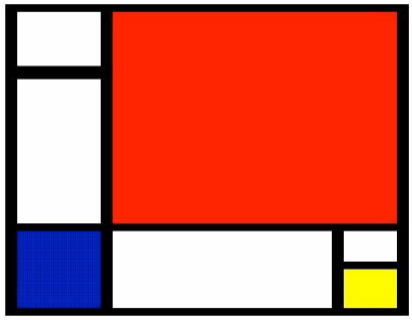
```
    // Calculate mark classification
    var mark = 55;

if (mark >= 70){
    console.log("This is a First");
}
else if (mark >= 60 && mark <= 69){
    console.log("This is a 2:1");
}
else if (mark >= 50 && mark <= 59){
    console.log("This is a 2:2");
}
else if (mark >= 40 && mark <= 49){
    console.log("This is a Third");
}
```

Run your code by loading the webpage in Google Chrome with the JavaScript console visible. This program using a set of conditions to test what classification a mark is and outputs this using the `console.log()` method.

Change the value for the mark variable to test whether each condition works as expected.

Task 2: Mondrian Generator



"*Piet Mondrian was a Dutch painter and an important contributor to the De Stijl art movement, which was founded by Theo van Doesburg. Despite being well-known, often-parodied, and even trivialized, Mondrian's paintings exhibit a complexity that belie their apparent simplicity. The non-representational paintings for which he is best known, consisting of rectangular forms of red, yellow, blue, or black, separated by thick, black, rectilinear lines, are actually the result of a stylistic evolution that occurred over the course of nearly thirty years, and which continued beyond that point to the end of his life.*"

https://en.wikipedia.org/wiki/Piet_Mondrian

Until now, you have been outputting the results of your programs to the JavaScript console. However, the console is restricted to showing unformatted text so the next step is to start to output the results of our code in the HTML pages itself. One method of doing this is to use the HTML5 `canvas` element, which "*is used to draw graphics, on the fly, on a web page*":

http://www.w3schools.com/html/html5_canvas.asp

The `canvas` element is simply an area on a webpage where you can draw graphics using JavaScript. In a new HTML page called **"week8_task2.html"** type the following "skeleton" code in the `body` of the page that will generate a very simple version of a Mondrian painting using a combination of functions and conditions. You were shown the completed code during the lecture.

Copy this code into a new HTML document taking care to think about what each line is doing.

mondrian_scaffold.html > html > body > script > draw

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Document</title>
</head>
<body>
<canvas id="sCanvas" width="550" height="400"></canvas>

<script>
    // Variables
    // Colours
    const blue = "#0000FF";
    const green = "#006600";
    const brown = "#543725";
    const red = "#FF0000";
    const yellow = "#FFFF00";
    const grey = "#666666";
    const white = "#FFFFFF";
    const black = "#000000";
    let colour; // global variable for use in functions
```

```
22
23      // Get the Canvas element with ID sCanvas that will be drawn on
24      const c = document.getElementById("sCanvas");
25      const ctx = c.getContext("2d");
26
27      // Draw a square with a specfied size, position and colour
28      function drawFilledShape(sWidth,sHeight, x,y,colour){
29          ctx.beginPath();
30          ctx.rect(x,y,sWidth,sHeight);
31          ctx.fillStyle = colour;
32          ctx.fill();
33          ctx.strokeStyle = black;
34          ctx.lineWidth = 5;
35          ctx.stroke();
36      }
37
38      function turnfNameIntoColour(fName){
39          if(fName.length > 6){
40              colour = white;
41          }
42          else{
43              colour = grey;
44          }
45          return colour;
46      }
47
48  function turnAgeIntoColour(age){
49
50          return colour;
51      }
52
53      function turnEyeColourIntoColour(eyeColour){
54
55          return colour;
56      }
57
58      function draw(){
59       let age = 53; //using let as will be taking user input later
60       let eyeColour = "blue"; // choose from: blue, green, brown
61       let fName = "Matt";
62
63      // Turn details into colours
64      const sColourTopLeft = turnAgeIntoColour(age);
65      const sColourBottomLeft = turnfNameIntoColour(fName);
66      const sColourRight = turnEyeColourIntoColour(eyeColour);
67
68      // Draw the coloured shapes
69      drawFilledShape(150, 150, 0, 0, sColourTopLeft);
70      drawFilledShape(150, 250, 0, 150, sColourBottomLeft);
71      drawFilledShape(400, 400, 150, 0, sColourRight);
72  }
73
74  draw();  //call the main draw function
75  </script>
76  </body>
77  </html>
```

At the moment this code will run but you will notice that two of the squares (`sColourBottomLeft and sColourRight`) are grey and one is black (`sColourTopLeft`).

Your task is to create a set of conditions in the functions turnEyeColourIntoColour() and turnAgeIntoColour() to change the colour of those squares depending on a value that is sent to the function.

To help you, you have been given the code (line 38) for the `turnfNameIntoColour()` function which has a String parameter, `fName`, and then checks to see whether the length of `fName` i.e. how many letters, is greater than 6 (`fName.length > 6`). If it is, then the function returns the colour as `white`, otherwise it returns the colour as `black`.

Once you have typed in the code, try and understand what each section does.

turnEyeColourIntoColour

For the `turnEyeColourIntoColour` function create a set of conditions that simply takes the `eyeColour` parameter and sets the colour variable to be the same as the value of the `eyeColour` parameter e.g.

```
if(eyeColour == "green"){
      colour = green;
}
```

do this for the colours green, blue and brown.

turnAgeIntoColour

For the `turnAgeIntoColour` function, create a set of conditions that takes the `age` parameter and if `age` is less than 30, the colour is `yellow`, otherwise the colour is `red`.

Once you have created these sets of conditions, change the values of the `age`, `eyeColour` and `fName` variables to your own details and see what your personalised Mondrian looks like.

Task 3: Mondrian Generator with user input

In the lecture you were shown an improved version of the Mondrian generator that allowed the user to enter their age, name and select eye colour from a drop-down menu.

Re-save your code from task2 ***as*** **"week8_task3.html"**

For task 3 you are to create a text box for the user to enter their name into and a button that when clicked will take their name and use it as a value for the variable fName in the code.

Steps you will need to carry out.

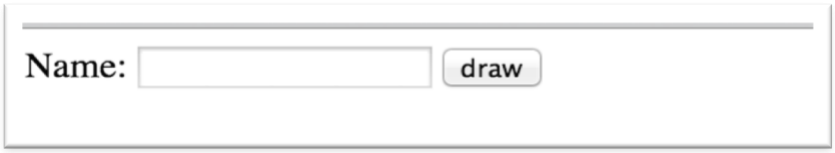
1. Comment out the function call draw() on line 74, we want to use a button to call the function, not have it run automatically.

```
72  }
73
74  //draw();  //call the main draw function
75  </script>
```

2. In the html create a text box with id ‘fName’ and a button that calls function draw() when clicked (you were given the code for this in the interactive area calculator in the previous lab session). These elements should be within the body of your HTML doc but before the <script> tags.
3. Remove the hard coded value assigned to variable fName (see line 61 below) and add the code necessary to get the value form the text box you created above in step 1. (This technique was explained again in todays lecture and given to you in the previous lab session).

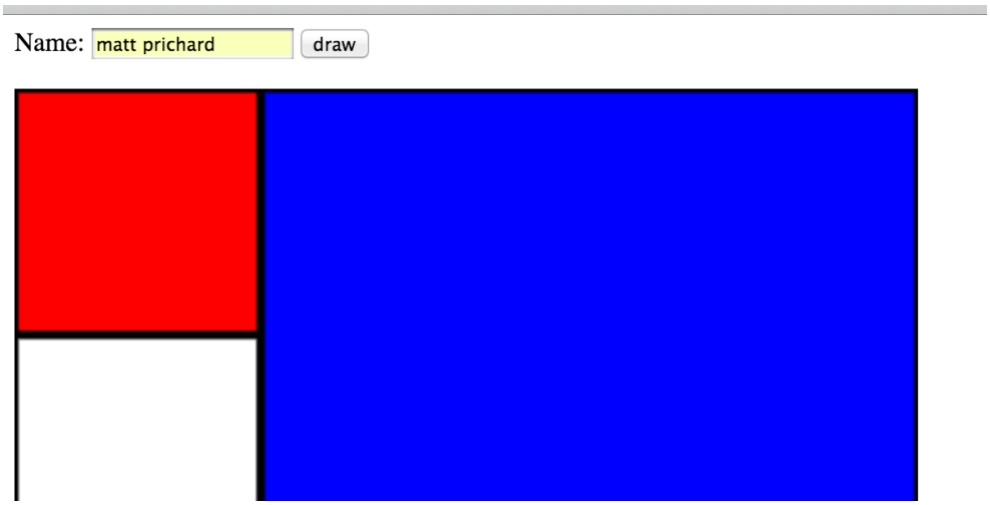
```
58      function draw(){
59        let age = 53; //using let as will be taking user input later
60        let eyeColour = "blue"; // choose from: blue, green, brown
61        let fName =
```

When you load the page you should see something like this:



Nothing is being drawn yet because the button click will call the draw function.

Enter your name and click the button, you should see something like this:

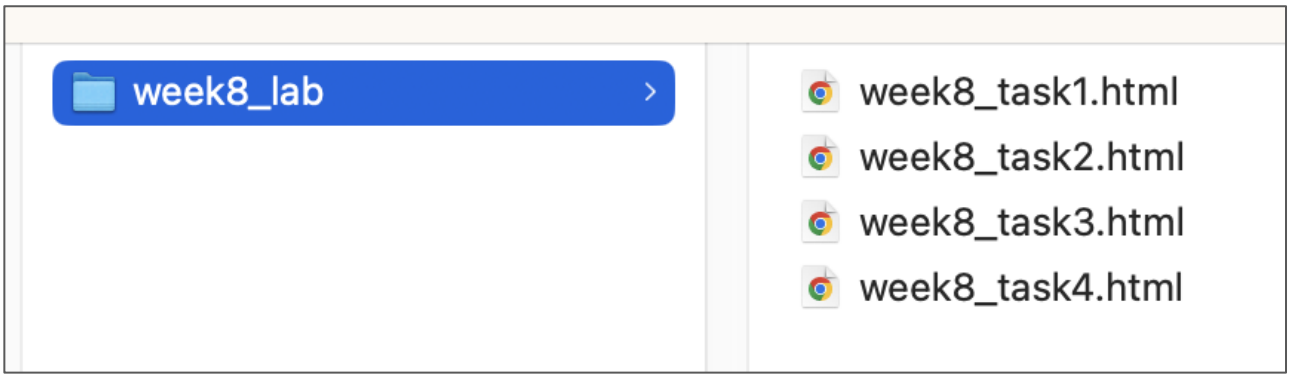


Task 4: Mondrian Generator with user input extended

The example I showed in the lecture had a text box for age and a drop down menu for eye colour. Save **week8_task3.html** as **week8_task4.html**

If you have got the code for the user's name and the draw button working see if you can implement age and eye colour too.

Questions to consider

1. Why have I used a drop-down list for eye colour and not just provided another text box?
2. In the example I showed in the lecture I had alert boxes popping up if I did not enter anything into the text boxes before clicking the button. This technique is known as *form validation* or *data input validation*. In the code I am asking questions, using if statements, such as "is this text box empty" or "is this data a number or not". If not display an alert box message and require the user to go back and try again.

Upload the following into the coursework upload system.